

AGRISPIKE – PRELIMINARY PATENT / PRIOR-ART RESEARCH REPORT

Intelligent Zone-Based Farming System • Preliminary Research Strategy

Purpose: Preliminary research for hackathon/project presentation. This report identifies similar published patent documents and highlights areas that may require further patentability analysis.

Important: This is a preliminary technical prior-art report, not a legal opinion and not a confirmation that AgriSpike is patentable.

1. AGRISPIKE - PROPOSED INVENTION

AgriSpike is an intelligent zone-based farming concept in which a farm is divided into zones. Smart Spike sensor nodes collect parameters such as soil moisture, temperature/humidity, and potentially soil nutrients and pH. Sensor data is transmitted through LoRa to a central master/gateway, forwarded to cloud storage, displayed on a dashboard, and used to support zone-specific irrigation through valves. Solar power and a battery can support outdoor sensor nodes.

Zone/plant sensors → Smart Spike node → LoRa master/gateway → cloud/database → dashboard → irrigation decision → valve/relay

2. PATENTABILITY REQUIREMENTS IN INDIA

According to the Indian Patent Office, an invention generally needs to be novel, involve an inventive step, be capable of industrial application, and not fall under the exclusions in Sections 3 and 4 of the Patents Act. The Patent Office defines an invention as a new product or process involving an inventive step and capable of industrial application.

Therefore, the broad statement "*smart irrigation using sensors and LoRa*" should not automatically be treated as a new invention. The specific technical implementation and its differences from existing prior art must be examined.

3. KEY SIMILAR PATENT / PRIOR-ART DOCUMENTS FOUND

Patent Document	Main Features Relevant to AgriSpike	Overlap Assessment
CN216874335U (CN113951115A family)	LoRa agricultural irrigation monitoring; solar power, battery; soil temperature/humidity, pH, EC, nutrients; environmental temperature/humidity & illumination; remote valve control.	High overlap with broad AgriSpike architecture.
CN118318713A	Multiple farmland sensors/nodes; soil moisture; temperature/humidity; meteorological data; LoRa node-to-gateway communication; cloud platform; automated water valve control.	High overlap with distributed smart irrigation.

4. DETAILED PRIOR-ART FINDINGS

A. Detailed Finding – CN216874335U

The document is titled *"Cotton field irrigation Internet of things monitoring system based on LoRa."* It describes a solar power supply, field information acquisition, meteorological monitoring, LoRa base station, valve terminal control, execution device, and IoT cloud platform. Its sensors include soil temperature/humidity, soil pH, soil EC, soil nutrients, environmental temperature/humidity, and illumination. It also describes remote valve control through the cloud platform.

Significance: This document is especially important because many individual components proposed for AgriSpike are already disclosed together in one system. Therefore, these individual features alone should not be presented to judges as completely new.

Source: Google Patents, CN216874335U. Priority date: 28 Nov 2021; publication: 5 Jul 2022. Original Assignee: Shihezi University.

B. Detailed Finding – CN118318713A

This document describes an intelligent irrigation system for high-standard farmland using multiple data acquisition sensors, nodes, intelligent water valves, a gateway, and a cloud management platform. It describes soil moisture, soil temperature/humidity, and meteorological monitoring, LoRa communication from nodes to a gateway, solar panel and battery power, and control of water valves. It also describes combining multiple data sources for irrigation decision-making.

Significance: This is another strong prior-art reference against a broad claim covering distributed agricultural sensors + LoRa + gateway + cloud + solar/battery + automated irrigation.

5. WHAT MAY NEED FURTHER NOVELTY INVESTIGATION

The following are not declared patentable; they are areas to investigate against prior art:

- Specific physical/mechanical Smart Spike architecture and how it is deployed in zones.
- A particular zone-identification and zone-wise control mechanism.
- A specific low-power wake → sense → transmit → sleep implementation, if it provides a measurable technical improvement.
- A specific power-management architecture combining solar charging, battery operation, and the node's duty cycle.
- A specific multi-parameter technical control method that produces a technical improvement, rather than merely a software rule.
- A particular integration of sensing, communication, and valve actuation that is technically different from the cited systems.

6. PRELIMINARY PATENTABILITY ASSESSMENT

Finding: The broad AgriSpike concept appears to have substantial prior art. Existing documents already disclose many of the same building blocks and combinations, including LoRa, agricultural sensors, solar/battery power, cloud monitoring, and automatic valve control.

However: This does not prove that every specific implementation of AgriSpike is unpatentable. Patentability depends on the exact claims, technical details, priority dates, complete prior-art search, and examination by a qualified patent professional.

Recommended Position Before Judges / Reviewers:

"Our preliminary prior-art study shows that the broad smart-irrigation architecture has existing patents. Therefore, we are focusing our novelty investigation on the specific Smart Spike architecture, zone-wise implementation, low-power operation, and technical integration rather than claiming that the general concept is entirely new."

7. SOURCES & REFERENCES

- Indian Patent Office - Basics of Patents: <https://ipindia.gov.in/basics-of-patents>
- Google Patents - CN216874335U: <https://patents.google.com/patent/CN216874335U/en>
- Google Patents - CN118318713A: <https://patents.google.com/patent/CN118318713A/en>
- WIPO PATENTSCOPE: <https://patentscope.wipo.int/search/en/search.jsf>